Machine learning practice questions

1. Consider a single data point $x = 2, y = 10$ and a linear model
$$\hat{y} = wx$$
The loss is
$$L(w) = \frac{1}{2}(wx - y)^2$$
If current weight is $w = 1$ and learning rate is $\eta = 0.1$:
a) Compute the gradient $\frac{\partial L}{\partial w}$
b) Perform **one step** of gradient descent to find the updated weight w_{new}
c) Explain whether a larger η would risk divergence.
Answer
a) Gradient, $\frac{\partial L}{\partial w} = (wx - y)x$
Initial gradient, $\frac{\partial L}{\partial w} = -16$
b) One update:
$$w_{new} = w - \eta \frac{\partial L}{\partial w} = 1 + 16 * 0.1 = 2.6$$
c) Yes. Can overshoot minimum and diverge/ oscillate

2. You collect the following data of building height vs wind load,

Height (m)	**Load (kN)**
10	18
20	31
30	45
40	62

You try the following models:
- Model A: linear $WL = a + bh$
- Model B: cubic $WL = c_0 + c_1h + c_2h^2 + c_3h^3$

a) Which model is more prone to overfitting? Why?
b) Explain how you would detect overfitting.

Answer

a) Model B - has more expressibility (ability to fit more functions). Therefore, can also fit the noise in the data and lead to overfitting.
b) Split data into training and validation data. Train the model for a few iterations (say 100) and find final loss on training and validation data. Overfitting is detected when training loss is very low but validation loss is high. We can use this to select optimal hyperparameters as well.

3. Suppose two features are used to predict material strength:

Predictor 1, x_1	**Predictor 2, x_2**	**Output, y**
2000	5	40
2100	7	38
2300	9	35

a) Compute mean and standard deviation of each predictor
b) Normalize the predictor values to zero mean and unit variance ($x_{norm} = \frac{x-\mu_x}{\sigma_x}$)
c) Explain why we need normalization

<u>Answer</u>

a) x_1

Mean of x_1: $\frac{(2000+2100+2300)}{3} = 2133.3$

Std deviation of x_1: $\sqrt{\frac{(2000-2133.33)^2+(2100-2133.3)^2+(2300-2133.3)^2}{3}} = 124.7$

x_2

Mean of x_2: $\frac{(5+7+9)}{3} = 7$

Std deviation of x_1: $\sqrt{\frac{(5-7)^2+(7-7)^2+(9-7)^2}{3}} = 1.6$

b) Normalizing predictors:

x_1	$\frac{x_1 - \mu_{x_1}}{\sigma_{x_1}}$	x_2	$\frac{x_2 - \mu_{x_2}}{\sigma_{x_2}}$
2000	-1.069	5	-1.225
2100	-0.267	7	0
2300	1.336	9	1.225

c) Normalization makes predictors comparable. This in turn stops Gradient Descent from being influenced by one predictor heavily due to large gradients in that predictor direction. Thus, optimization becomes easier

4. You are given the following data for predicting concrete compressive strength from water–cement ratio:

Water-cement ratio, x	**Strength, y**
0.5	42
0.6	38

You assumed linear model, $y = a + bx + \epsilon$

a) Write design matrix $\boldsymbol{A}$ and observation matrix $\boldsymbol{y}$
b) Compute least squares estimate $\hat{\beta} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{y}$
c) Find mean squared error loss using these optimal parameters

<u>Answer</u>

a) Design matrix

$$A = \begin{bmatrix} 1 & 0.5 \\ 1 & 0.6 \end{bmatrix}, y = \begin{bmatrix} 42 \\ 38 \end{bmatrix}$$

b) $\hat{\beta}$

$$\hat{\beta} = (\boldsymbol{A}^T\boldsymbol{A})^{-1}\boldsymbol{A}^T\boldsymbol{y} = \begin{bmatrix} 62 \\ -40 \end{bmatrix}$$

Thus, we have $y = 62 - 40x + \epsilon$
Notice how coefficient of x is negative!

c) $MSE = \frac{1}{2}[(42 - 62 + 40 * 0.5)^2 + (38 - 62 + 40 * 0.6)^2] = 0$

5. You want to classify a new soil sample based on density and moisture.

Density, x_1	**Moisture, x_2**	**Soil type**
1.0	6	Clay
1.5	4	Sand
2.0	8	Clay
2.5	5	Sand

a) What is the type of task you are trying to solve (classification, regression or clustering) ?
b) You now are given a new point, density = 1.7 and moisture = 6. Compute Euclidean distance to all points.
c) What is the soil type if you use KNN with k=1?
d) What is the soil type if you use KNN with k=3?

<u>Answer</u>

a) Since y is discrete (Clay or Sand), classification task
b) Euclidean Distance,

$$d = \sqrt{(x_{1new} - x_{1i})^2 + (x_{2new} - x_{2i})^2}$$

To point 1: $d = \sqrt{(1.7 - 1)^2 + (6 - 6)^2} = 0.70$
To point 2: $d = \sqrt{(1.7 - 1.5)^2 + (6 - 4)^2} = 2.00$
To point 3: $d = \sqrt{(1.7 - 2)^2 + (6 - 8)^2} = 2.02$
To point 4: $d = \sqrt{(1.7 - 2.5)^2 + (6 - 5)^2} = 1.28$

c) Soil type with k=1:
1 nearest neighbor are: point 1 (smallest d)
Therefore, prediction is: y_1= Clay

d) Soil type with k=3:
3 nearest neighbors are: point 1, point 4 and point 2 (smallest d)
Therefore, prediction is: mean(y_1, y_4, y_2)
Let us assign $Clay = 0, Sand = 1$
mean(y_1, y_4, y_2) $= mean(0,1,0) = 0.33 < 0.5$
Therefore, Clay!

Alternatively, we can use mode(y_1, y_4, y_2) $= mode(Clay, Sand, Clay) = Clay$

6. You have a dataset of 120 samples. You split it as: 70% training, 15% validation, 15% testing.
a) How many samples go into each set?
b) Explain the purpose of the validation set.

c) What happens if the test set is used to tune hyperparameters?

<u>Answer</u>

a) Training: $120 \times 0.7 = 84$
   Validation: $120 \times 0.15 = 18$
   Testing: $120 \times 0.15 = 18$
b) Purpose of validation set: Tune the model by selecting hyperparameters that lead to lowest validation error. Thereby, we also avoid overfitting
c) If test set is used to tune hyperparameters, it is like leaking the question paper before an exam. The reported error of the model becomes optimistically low.

7. You observe the following validation errors after training the linear regression model with different learning rates:

Learning rate	**Validation Error**
0.001	12
0.01	6
0.1	1200

a) Do these validation errors make sense to you? Why or why not?
b) What is the learning rate you would use for training?
c) What do you classify learning rate as? Model parameter or hyperparameter? Why?

<u>Answer</u>

a) Yes. Low η leads to very slow learning and thus convergence. This leads to undertraining. High η leads to overshooting from the optimal point and thus divergence. Optimal η leads to optimal learning.
b) $\eta = 0.01$ since it leads to lowest validation error. In practice you might try a small grid around 0.01 (e.g., 0.005, 0.02) to further select the best learning rate.
c) Learning rate is a **hyperparameter**. It is not learned from data during model fitting but set by the us to make optimization algorithm work.

8. You attempt a polynomial regression of degree d to predict load from displacement. The measured errors are:

Degree, d	**Train error**	**Validation error**
1	12	14
2	6	7
5	1	20

a) Which model suffers from high bias?
b) Which model suffers from high variance?
c) Which model should you pick and why?
d) You get a second set of data from experiments. In which model will the parameters be very different from the earlier data?

<u>Answer</u>

a) Degree 1 has high bias as indicated by high training error. The data contains the complicated patterns but the model has been given limited expressibility (think of it as number of functions the model can fit). No matter how much you train some patterns cannot be learnt due to lower model expressibility (degree).
b) Degree 5 has variance as indicated by very low training error but very high validation error. There were more model parameters than required (due to high degree) and thus the model has started learning (fitting) noise as well. This leads to overfitting.
c) Degree 2 is the optimal. Low training error and comparable validation error.
d) Degree 5 fits the noise as well. Since different set of data will have different noises, it will leads to large changes in the estimated model parameters.

9. Given data:
$x = [1,2,3]$ and $y = [2,2.5,3.5]$
a) Compute closed from Least Squares estimate assuming a model of the form $y = bx$
b) If initial guess is $b_0 = 0$, find the gradient descent update after 1 iteration with learning rate = 0.1
c) Why do you think we would use gradient descent even for least-squares estimate for large scale data.

Answer

a) Writing out equations, we have

$$\begin{bmatrix} 2 \\ 2.5 \\ 3.5 \end{bmatrix} = b \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Multiply with x^T

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 2.5 \\ 3.5 \end{bmatrix} = b \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$17.5 = b(14)$$
$$b = 1.25$$

b) Gradient:

$$\frac{\partial L}{\partial b} = \sum (bx_i - y_i)x_i$$

Initially, at $b_0 = 0$

$$\frac{\partial L}{\partial b} = \sum (0 - y_i)x_i = -\sum x_i y_i = -17.5$$

Gradient descent update:

$$b_1 = b_0 - \eta \frac{\partial L}{\partial b} = 0 + 17.5 * 0.1 = 1.75$$

Notice how we have overshot the analytical b because in this case $\eta = 0.1$ is high.

c) Closed form requires inverting matrices which is extremely costly for large datasets. Many times closed form expressions are not available as well. Gradient descent methods scale much better.

10. You want to predict compressive strength of concrete by using the water-cement ratio.

w/c ratio	Strength (MPa)
0.4	55
0.52	42
0.6	36
0.65	33

You are now asked to predict the strength of concrete with water-cement ratio of 0.58. You decide to use kNN regression.

a) What is your estimate for k=1?
b) What is your estimate for k=2?
c) Which kNN tends to overfit more? k=1 or k=5? Why?

Answer

a) Euclidean distances are:
Point 1: $|0.4 - 0.58| = 0.18$
Point 2: $|0.52 - 0.58| = 0.06$
Point 3: $|0.6 - 0.58| = 0.02$
Point 4: $|0.65 - 0.58| = 0.07$

Thus kNN prediction for water-cement ratio of 0.58 with k=1: Point 3 is nearest

$$\hat{y} = mean\left(y_{P_1}, y_{P_2}, .., y_{P_k}\right) = mean(y_3) = 36\ MPa$$

b) Thus kNN prediction for water-cement ratio of 0.58 with k=2: Point 3 is nearest

$$\hat{y} = mean\left(y_{P_1}, y_{P_2}, .., y_{P_k}\right) = mean(y_3, y_2) = 39\ MPa$$

c) k=1 overfits more single predictions follow single nearest neighbor causing it to learn noise as well. Higher k follows a bunch of neighbors and therefore leads to averaging which reduces the noise, leading to smoother prediction.

11. You are fitting polynomial models to the following dataset:

Example Number	x	y
1	0	1
2	1	2
3	2	2
4	3	3

You decide to use quadratic model: $y = c_0 + c_1 x + c_2 x^2$. You need some data for validation, so you decide to keep one data point out. You try out two models. In the first model, you use the 1st, 2nd and 3rd examples for training and 4th for validation. In the second model, you use the 2nd , 3rd and 4th for training and 1st for validation. Find the least squares estimate for parameters of these two models. Explain why they are so different.

Answer

Model assumption:

$$y = c_0 + c_1 x + c_2 x^2$$

The design matrix is thus

$$\boldsymbol{y} = \boldsymbol{Ax} + \epsilon$$

When we use the first three examples for training, we get:

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} + \epsilon$$

LS estimate:

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = (A^T A)^{-1} A^T y = \begin{bmatrix} 1 \\ 1.5 \\ -0.5 \end{bmatrix}$$

When we use the last three examples for training, we get:

$$\begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} + \epsilon$$

LS estimate:

$$\begin{bmatrix} c_0 \\ c_1 \\ c_2 \end{bmatrix} = (A^T A)^{-1} A^T y = \begin{bmatrix} 3 \\ -1.5 \\ 0.5 \end{bmatrix}$$

In this case, expressibility is too much leading to model fitting noise as well. Since every data point will have different noise, when we created two datasets each with different noises. The two models thus fit these noises perfectly, leading to vastly different set of parameters. This is the case of classic overfitting we should avoid. Note that the estimates if we use all 4 examples will be more realistic are, $c_0 = 1.1, c_1 = 0.6\ and\ c_2 = 0$. This also shows quadratic term was not really needed!